

$^3\text{He}(n, \cdot)^4\text{He}$ capture cross sections and the IPC pair yield

The measured M1+E1 radiative capture, the γ -dark E0, and the e^+e^- yield that follows
from $\sigma \times (\text{pair fraction})$

Dylan Neff

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*Internal note — informal. Drafted by Claude (Anthropic AI assistant) under D. Neff's direction;
physics, numbers and text reviewed by D. Neff.*

Part 0 — Summary

The capture cross section factorises as $\sigma = \sigma_{\text{form}} \times (\Gamma_X/\Gamma_{\text{tot}})$ — **form the $^4\text{He}^*$ state, then radiate to the ground state rather than break up.** Two consequences drive this note:

- **The photon-emitting capture (M1+E1) is MEASURED** — it is the ENDF $^3\text{He}(n, \gamma)^4\text{He}$ cross section (55 μb at thermal, the classic value), with the M1-dominated $1/v$ at low E_n and the E1/GDR rise toward MeV. **Only the γ -dark E0 is estimated:** $\sigma_{\text{E0}} = f \sigma_{\text{M1}}$, $f \sim 10^{-3}$ – 10^{-2} , sitting 4–5 decades below.
- **The IPC (e^+e^-) yield is just $\sigma \times (\text{pair fraction})$:** $\times \alpha_{\text{IPC}} \approx 3.5 \times 10^{-3}$ for M1+E1 (only the internal-pair tail of the photon makes a pair), $\times 1$ for E0 (γ -dark $\Rightarrow$ 100 % pairs). Multiplying the measured curve *down* by α_{IPC} while leaving E0 alone lifts E0 by $1/\alpha_{\text{IPC}} \approx 285$: **the two pair yields become COMPARABLE at thermal/sub-keV** ($\text{E0/M1E1} = f/\alpha_{\text{IPC}} \approx 0.3$ – 3), while M1+E1 wins at MeV.
- **The dominant uncertainty is f** (~ 1 decade); the measured $\sigma_{n\gamma}$ is firm and α_{IPC} contributes only $\sim \times 1.8$.

1 The cross section is a product: form $\times$ radiate

For capture through a compound $^4\text{He}^*$ state of spin J ,

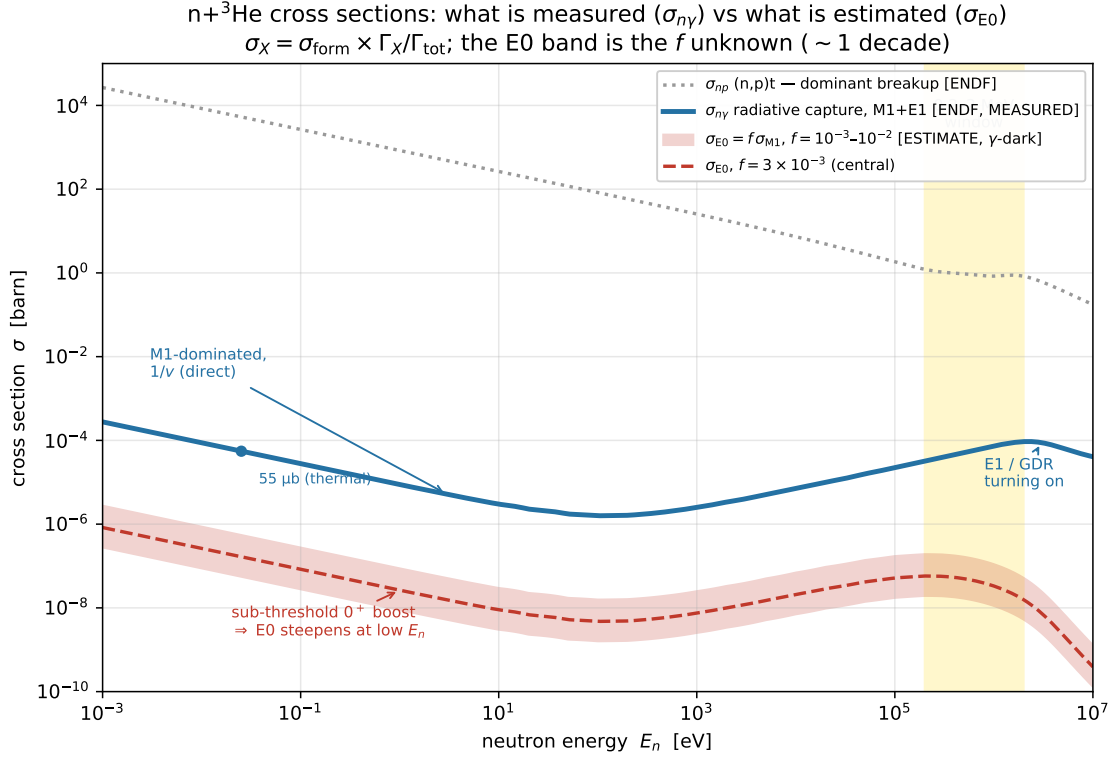
$$\sigma(n, X) = \underbrace{\frac{\pi}{k^2} g_J \frac{\Gamma_n(E) \Gamma_{\text{tot}}}{(E - E_r)^2 + (\Gamma_{\text{tot}}/2)^2}}_{\sigma_{\text{form}} \quad (\text{form the state})} \times \underbrace{\frac{\Gamma_X}{\Gamma_{\text{tot}}}}_{(\text{radiate as } X)} .$$

The total width is almost all **breakup** ($\Gamma_{\text{tot}} \approx \Gamma_n + \Gamma_p$, the (n,p)t channel), so the radiative branch is a $\sim 10^{-8}$ sliver — that is why capture is rare. Per channel, $\sigma_{\text{M1}} = \sigma_{\text{form}}(1^+) \Gamma_{\text{M1}}/\Gamma_{\text{tot}}$ and $\sigma_{\text{E0}} = \sigma_{\text{form}}(0^+) \Gamma_{\text{E0}}/\Gamma_{\text{tot}}$, so

$$f \equiv \frac{\sigma_{\text{E0}}}{\sigma_{\text{M1}}} = \underbrace{\frac{\sigma_{\text{form}}(0^+)}{\sigma_{\text{form}}(1^+)}}_{\sim 1-2 \text{ (rough)}} \times \underbrace{\frac{\Gamma_{\text{E0}}}{\Gamma_{\text{M1}}}}_{\text{the unknown}} .$$

(^4He capture is mostly *direct*, so “form then decay” is not a literal two-step process — but the cross section still separates into a kinematic/penetrability factor $\times$ an EM-strength factor, which is what matters here. The formation side is detailed in `docs/e0_branch/doorway_states_note.md`.)

2 The capture cross sections: measured vs estimated



Reading the figure:

- σ_{np} (**grey, dotted**) — the dominant (n,p)t breakup, 5330 b at thermal. This *is* Γ_{tot} , and the reference for “rare.”
- $\sigma_{n\gamma}$ (**blue, MEASURED**) — total radiative capture (M1+E1), ENDF/B-VIII.0, 55 μb at thermal. The $1/v$ falloff is the M1-dominated direct capture; the rise toward MeV is the E1/giant-dipole strength (the $T = 1$ 1^-) turning on. **No estimation enters this curve.**
- σ_{E0} (**red band, ESTIMATE**) — the γ -dark E0 capture, $\sigma_{E0} = f \sigma_{M1}$ with the sub-threshold 0^+ boost steepening it toward low E_n . The band is $f = 10^{-3}$ – 10^{-2} ; $\sigma_{E0}(\text{thermal}) \approx 0.055$ – $0.55 \mu\text{b}$. Invisible to $\sigma_{n\gamma}$ (it emits no photon).

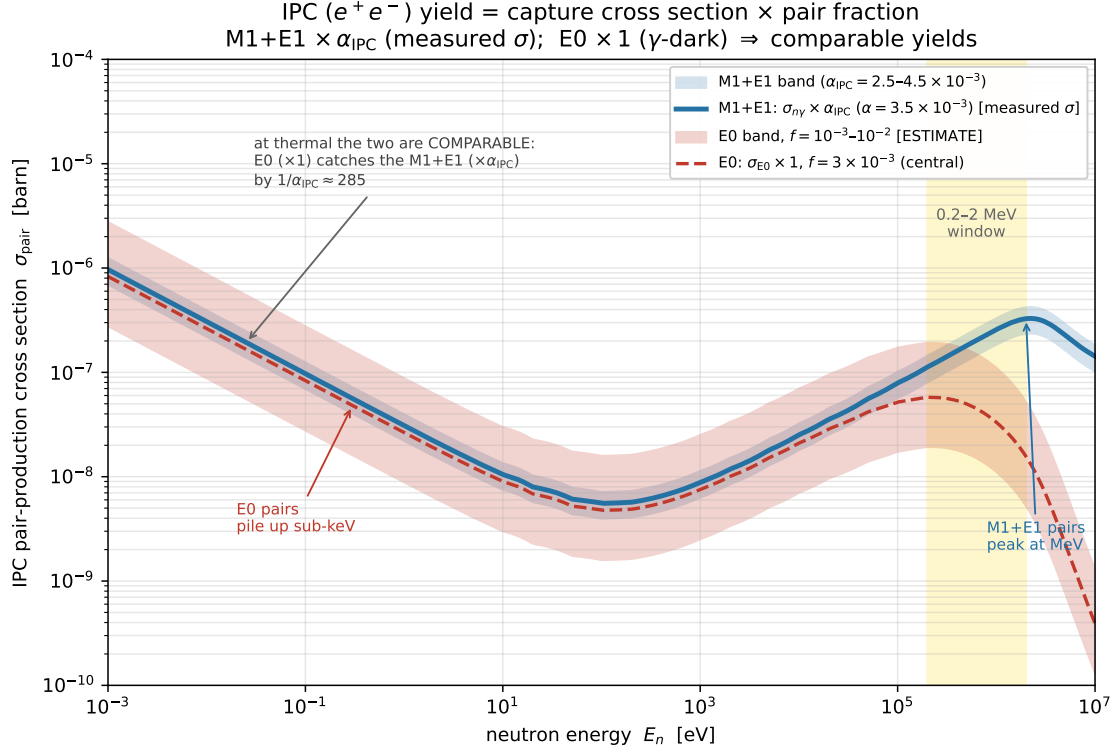
The honest split is stark: a measured photon capture, and a γ -dark E0 capture 4–5 decades below whose absolute height is the open f .

3 The IPC pair yield: $\sigma \times (\text{pair fraction})$

The e^+e^- yield is each cross section times its pair fraction — $\times \alpha_{\text{IPC}}$ for the photon channel (only the internal-pair tail of the γ becomes a pair), $\times 1$ for E0 (it makes nothing *but* pairs):

$$\sigma_{\text{pair}}(\text{M1} + \text{E1}) = \sigma_{n\gamma} \times \alpha_{\text{IPC}}, \quad \sigma_{\text{pair}}(\text{E0}) = \sigma_{E0} \times 1.$$

$\alpha_{\text{IPC}} \approx 3.5 \times 10^{-3}$ at 20.6 MeV (anchored to the measured ¹²C/⁸Be high-energy M1 IPC; band 2.5 – 4.5×10^{-3} ; Alberto’s table uses 2.1×10^{-3}).



The result. Multiplying the measured curve *down* by α_{IPC} while leaving E0 *unchanged* lifts E0 by $1/\alpha_{\text{IPC}} \approx 285$ relative to M1+E1. The 4–5-decade gap of the capture figure **collapses**: the two pair yields are **comparable at thermal** (at $E_n = 25$ meV, $\sigma_{\text{pair}}^{\text{M1+E1}} = 0.19 \mu\text{b}$ vs $\sigma_{\text{pair}}^{\text{E0}} = 0.055\text{--}0.55 \mu\text{b}$, i.e. $\text{E0/M1E1} = f/\alpha_{\text{IPC}} \approx 0.3\text{--}3$). They then split by energy — **E0 pairs pile up sub-keV, M1+E1 pairs peak at MeV** — so the γ -dark channel matters precisely where the ordinary one is dead.

4 X17 production, and the two scenarios

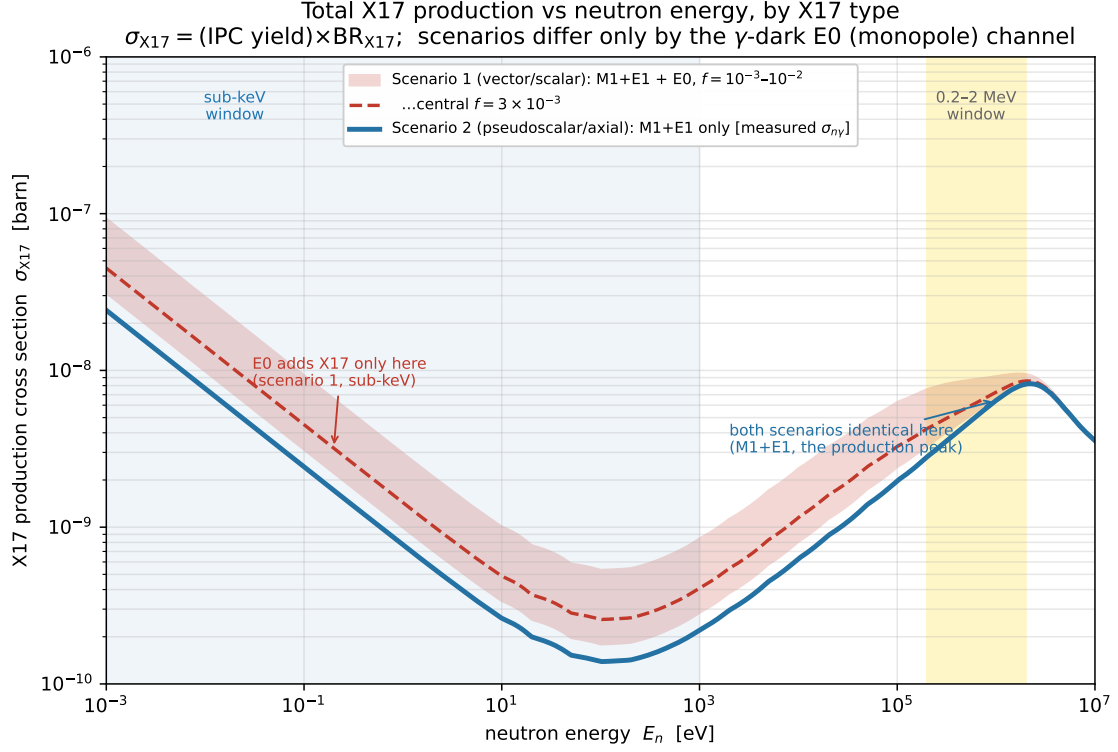
X17 is produced as a sub-branch of the same transitions: $\sigma_{\text{X17}} = (\text{IPC yield}) \times \text{BR}_{\text{X17}}$ with $\text{BR}_{\text{X17}} = 0.025$ (X17 per e^+e^- pair, the project benchmark). Which *modes* contribute depends on the X17 quantum numbers, through one decisive rule: a $0^+ \rightarrow 0^+$ (E0/monopole) transition can emit a massive boson only if its J^P allows $0^+ \rightarrow 0^+ + X$ —

X17	J^P	couples to E0?
vector (dark photon)	1^-	yes
scalar	0^+	yes
pseudoscalar	0^-	no
axial-vector	1^+	no

so the total X17 production splits into two scenarios, differing *only* by the γ -dark E0 channel:

- **Scenario 1 (vector or scalar):** M1+E1 + E0.
- **Scenario 2 (pseudoscalar or axial):** M1+E1 only.

(At the strict per-multipole level the M1/E1 couplings are also type-dependent; we take “X17 rides the measured M1+E1 strength” as the working approximation, so the clean scenario difference is the E0 piece.)



The two scenarios are **identical at the MeV production peak** (all M1+E1) and diverge only in the sub-keV region, where scenario 1 gets the E0 band.

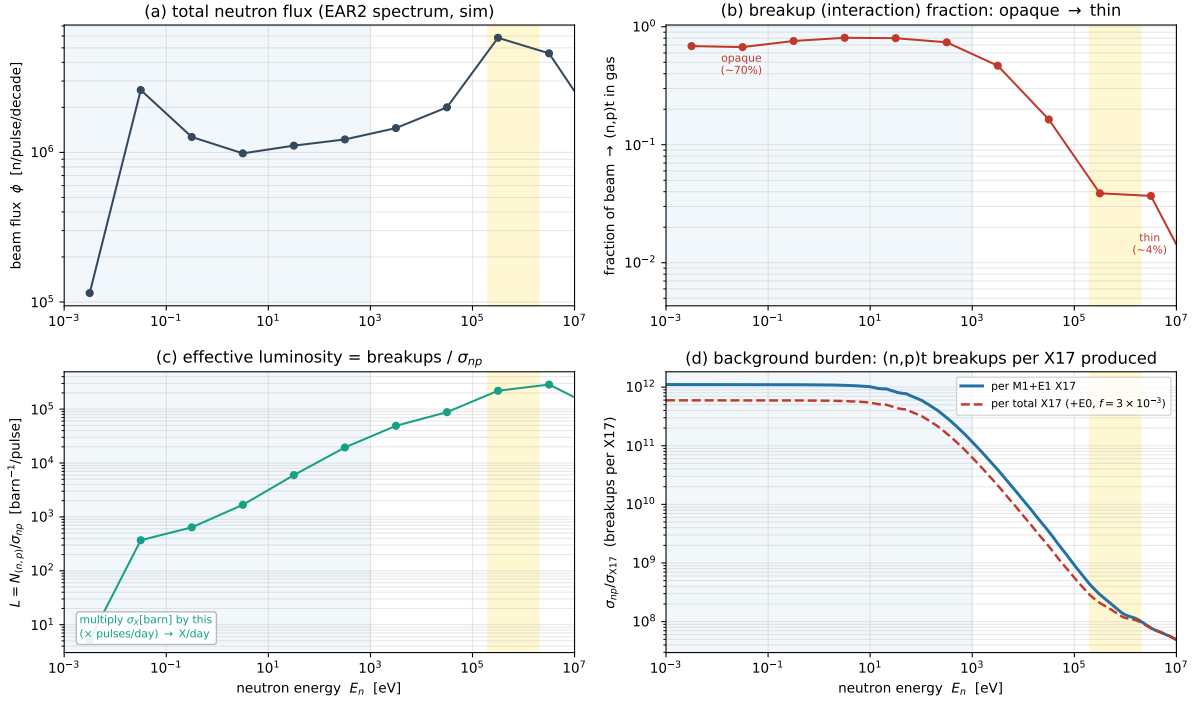
5 From cross section to statistics

To turn a cross section into a rate we fold in the simulated beam:

$$\frac{N_X}{\text{day}}(E_n) = \underbrace{\phi(E_n)}_{\text{flux}} \times \underbrace{f_{\text{breakup}}(E_n)}_{\text{fraction of beam interacting in gas}} \times \underbrace{\frac{\sigma_X(E_n)}{\sigma_{np}(E_n)}}_{\text{X per breakup}} \times \text{pulses/day},$$

equivalently $N_X = N_{(n,p)} \times \sigma_X / \sigma_{np}$ — **the (n,p)t breakup, counted directly in Geant4, is the luminosity monitor.** The clean part: even where the gas is *opaque* (sub-keV), the flux attenuation along the column is set by the dominant (n,p) absorption, so it multiplies *both* channels by the same $(1 - e^{-\tau_{np}})$ and **cancels in the ratio** — $N_X = N_{(n,p)} \sigma_X / \sigma_{np}$ holds exactly. (For M1+E1 we instead use the direct Geant4 (n, γ) count, which bakes in flux/geometry/opacity already; E0, absent from Geant4, is the one obtained via the ratio.)

From cross section to statistics: $N_X/\text{day} = \phi \times f_{\text{breakup}} \times (\sigma_X/\sigma_{np}) \times \text{pulses/day}$ (gold = 0.2–2 MeV window, blue = sub-keV)



The four factors, vs energy: **(a)** the EAR2 flux ϕ ; **(b)** the breakup fraction, $\sim 70\%$ (opaque) sub-keV falling to $\sim 4\%$ (thin) at MeV; **(c)** the effective luminosity $L = N_{(n,p)}/\sigma_{np}$ — multiply σ_X [barn] by this ($\times$ pulses/day) to get X/day; **(d)** the background burden, (n,p)t breakups per X17 produced, which falls from $\sim 10^{12}$ in the sub-keV to $\sim 10^8$ at MeV. Panel (d) is the quantitative reason the MeV window is favourable: $\sim 10^4 \times$ fewer breakups per signal than sub-keV. (The E0 channel, scenario 1, lowers the burden at low E_n — red dashed — but only into the $\sim 10^{11}$ range.)

The daily statistics (the number we want)

Folding the measured per-decade campaign flux (mev_rates.json), the **X17 produced per day** in the two windows:

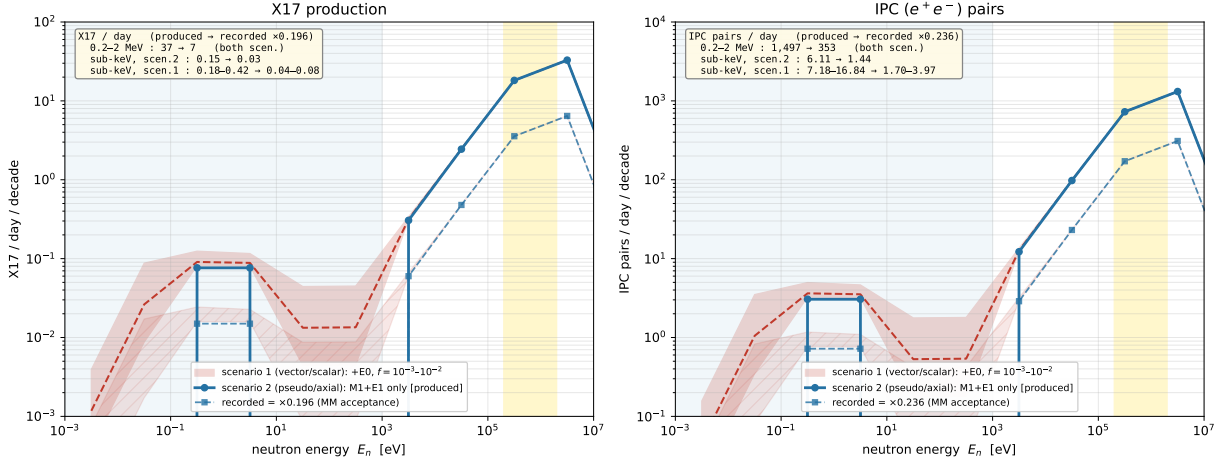
X17 produced / day	Scenario 2 (M1+E1 only)	Scenario 1 (vector/scalar) M1+E1 + E0, $f = 10^{-3}$ – 10^{-2}
high-energy window (0.2–2 MeV)	37.4	37.4 (E0 adds < 0.004)
full sub-keV window ($E_n < 1$ keV)	0.15	0.18–0.42

(Anchored $\alpha_{\text{IPC}} = 3.5 \times 10^{-3}$; the high-energy row is 22.5/day with the table $\alpha_{\text{IPC}} = 2.1 \times 10^{-3}$ — the committed headline. E0 alone contributes 0.03–0.27 X17/day in the sub-keV and < 0.004 /day in the high-energy window.)

What this says. (i) The high-energy window carries the X17 rate — ~ 22 – 37 produced/day — and is **identical in both scenarios**: E0 is irrelevant there, so the baseline measurement does not depend on the X17 type. (ii) The sub-keV window is **starved either way**: even the most favourable case (vector/scalar X17, $f = 10^{-2}$) gives only ~ 0.4 X17/day, $\sim 100 \times$ below the high-energy window; the E0 channel at best ~ 2 – $3 \times$ the (tiny) sub-keV M1+E1 rate. So a vector/scalar X17 would add a *distinct sub-keV signature*, but not significant *statistics*. (iii) These are *produced*; recorded $\approx \times 0.196$ (MM acceptance), and the sub-keV window additionally requires a thermal trigger (the 0.2–2 MeV window is already in the 10 μ s flash readout).

Everything folded: per day per decade

Folded result: X17 and IPC per day per decade, produced and recorded ($\times$ acceptance)
gold = 0.2–2 MeV window, blue = sub-keV; sub-keV recording also needs a thermal trigger (not applied here)



The full fold-down, per decade: X17 (left) and IPC pairs (right), *produced* and *recorded* ($\times$ the MM-double geometric acceptance, 0.196 X17 / 0.236 IPC). Scenario 2 (blue) is M1+E1 only; scenario 1 (red band) adds E0 over $f = 10^{-3}$ – 10^{-2} . The MeV peak (gold window) carries the rate and is scenario-independent; the sub-keV (blue) is where scenario 1's E0 lifts an otherwise-dead region — but to small absolute numbers. The boxed sums are the window totals (produced $\rightarrow$ recorded). (*Acceptance only; the sub-keV recording additionally needs a thermal trigger, and these use at-rest acceptance — the E_n -boost correction is pending.*)

6 Uncertainties

piece	size of uncertainty	how to shrink
$\sigma_{n\gamma}$ (M1+E1 capture)	small (evaluated data)	— (measured)
α_{IPC} (pair fraction, M1+E1)	$\sim \times 1.8$ (2.5 – 4.5×10^{-3})	high-energy IPC theory + ^{12}C anchor
$\sigma_{\text{form}}(0^+)/\sigma_{\text{form}}(1^+)$	$\sim \times 2$ – 3	^4He R-matrix reduced widths
$f = \sigma_{E0}/\sigma_{M1}$	~ 1 decade (10^{-3} – 10^{-2})	^4He R-matrix + (e, e') monopole norm.

f **dominates by far**, so the E0 pair yield should be quoted as an order-of-magnitude *band*, not a central value (as drawn). The formation ratio and α_{IPC} are sub-dominant and, for f , are already bundled into it. The one calculation that turns the red band into a line is a ^4He multi-level R-matrix (Hale's evaluated system) for the 1S_0 $n + ^3\text{He} \rightarrow \text{g.s.}$ E0 capture, normalised by the measured $^4\text{He}(e, e')$ monopole strength.

Reproducing this note

Figures: `scripts/make_e0_cross_section.fig.py` (capture cross sections, $\rightarrow$ `fig_e0_xsec`), `scripts/make_e0_ipc_yield.fig.py` (pair yield, $\rightarrow$ `fig_e0_ipcyield`), and `scripts/make_e0_x17_production.fig.py` (X17 production + the window daily statistics, $\rightarrow$ `fig_e0_x17`), and `scripts/make_e0_luminosity.fig.py` (the flux/breakup/luminosity bridge, $\rightarrow$ `fig_e0_lumi`), and `scripts/make_e0_folded.fig.py` (the folded per-day-per-decade result, $\rightarrow$ `fig_e0_folded`); all read `data/He3.h5` (ENDF/B-VIII.0) and `analysis/mev/mev_rates.json`. The formation physics behind σ_{form} : `docs/e0_branch/doorway_states.note.md`. The recorded-rate version (flux $\times$ acceptance $\times$ trigger window): `docs/report/e0_final_metric.note.tex`.